

Yassine Zouari

Machine learning Engineer

(+1) 502 698 0307 | y0zoua01@louisville.edu | [linkedin](#) |

EDUCATION

University of Louisville

Master Degree in computer science and Engineering

Higher school of Communication of Tunis(Sup'COM)

Computer Science Engineering

Preparatory Institute for Engineering Studies

Ranked 48 out of 1250 in National Entrance Examination to the Engineering Cycle

Louisville ,Ky , USA

Jan. 2025 – Present

Tunis, Tunisia

Sep. 2021 – Jul. 2024

Sfax, Tunisia

Sep. 2019 – Jul. 2021

EXPERIENCE

Graduate Research assistant

Multimedia research Lab at University of louisville

Jan 2025 – Present

Uofl, usa

- Conduct research on multimodal object detection using RGB and thermal (IR) imagery
- Designed and implemented a pixel-level fusion module based on spatial and channel attention mechanisms
- Developed a custom attention-based fusion network in PyTorch to adaptively weight RGB and IR features
- Integrated the fusion module before YOLO detection heads to enhance feature representation
- Evaluated models using mAP, precision-recall curves, ROC, and compare the fusion-based detection against single-modality and other existing fusion methods
- Collaborate with advisor on experimental design and result analysis

Graduation Internship – Automatic Target Detection

Multimedia research Lab at University of louisville

Feb 2024 – July 2024

uofl, USA

- Adapt and optimize multiple YOLO algorithms to detect Automatic targets in cluttered RGB and infrared images
- Experiment with standard methods to improve performance, such as data augmentation, data processing, regularization, and hyperparameter tuning
- Compare and visualize the performance of the object detectors using standard tools such as ROC curves, mAP, AUC
- Build a decision fusion method that combines detections from multiple object detectors and both modalities
- Analyze the performance of the fusion and compare it to the results of each detector

NIP Summer Internship

IP-Label Company

June 2023 – Aug 2023

Tunis, Tunisia

- Design and develop a chatbot capable of answering questions from users on technical and functional topics based on a pre-existing documentary database
- **Keywords:** NLP, Transformers, Python, Pytorch, Machine learning, RASA, Nlu, llama2, embeddings

PROJECTS

Tutored Project: Fire and Smoke Detection

Sep 2023 – Jan 2024

- Designed and implemented a Fire and smoke detection system utilizing RGB Imagery, traditional feature extraction techniques, and machine learning algorithms
- **Keywords:** HOG, Feature extraction, Image processing, Feature Fusion, SVM, RandomForest classifier, Decision Fusion, AdaBoost

TECHNICAL SKILLS

Data Science, Deep Learning Computer Vision, Feature Engineering Keras, Machine Learning, Matplotlib
Natural language processing, Numpy, OpenCV, Pandas PyTorch, Seaborn, Tensorflow, Python, C/C++, SQL